

prodDisruption

AI-Driven Flight Disruption Recovery

A multi-tenant, AI-powered platform that re-accommodates and notifies disrupted passengers — personalised to each passenger's value. Validated end-to-end on a production-aligned Microsoft Azure architecture.

Same flight, different answer

Student → ₹4,200 economy · High-value → ₹18,500 business, automatically

Whole flight, one AI pass

All impacted passengers recovered · distinct seats · 0 manual escalations

Compute-once, serve-many

Precomputed recovery served in < 1s (measured 0.26s)

Disruption is costly, manual and impersonal

- **Flight disruption costs the travel industry an estimated US\$60bn a year — about 8% of industry revenue.** *(Industry: Amadeus/T2RL, 2016)*
- European air-traffic-control delays alone cost €16.1bn (2015–Oct 2025), affecting ~1 billion passengers. *(Industry: IATA)*
- When a flight cancels, agents must re-accommodate hundreds of passengers fast — today this is manual, slow and inconsistent.
- Generic rebooking ignores passenger value: a student and a high-value flyer get the same treatment.
- Disruption spreads 'virally' — one cancellation cascades into call-centre surges and CX / NPS damage. *(Industry: Amadeus/T2RL)*

THE COST OF DISRUPTION

\$60bn

per year, travel industry (~8% of revenue)

€16.1bn

European ATC delays, 2015–2025

~1bn

passengers affected by those delays

Personalised recovery at machine speed

- Automate the re-accommodate-or-notify decision for every passenger — tailored to their persona and value.
- Agentic AI is reshaping travel operations; faster, consistent disruption handling lifts satisfaction and deflects contact-centre load. *(Industry: McKinsey — Agentic AI in travel)*
- **Proof point: an Asian carrier using LLM-based CX saw its customer-satisfaction index rise 15% and contact-per-passenger fall 30%. *(Industry: McKinsey)***
- The prize: lower handling cost per disruption, faster recovery, consistent personalisation, and a retained customer.
- **AIONOS has built and validated the core capability — the opportunity is to take it to live airline operations.**

PROVEN CX UPLIFT FROM AI

+15%

customer-satisfaction index

−30%

contact-per-passenger

Source: McKinsey

An Azure-native, multi-tenant AI platform

- Fetches the disruption event, reads the passenger's persona from the airline's customer-data platform, then re-accommodates (cancellation) or notifies (delay).
- Multi-tenant by design: each airline's APIs, AI agents, prompts and templates are isolated and configurable without redeployment.
- Validated today against high-fidelity mock airline / CDP / disruption providers; one tenant fully wired — additional tenants are configuration-only onboarding.
- Personas supported: Student · High Spender · Doctor · Defence · General.

BUILT NATIVELY ON MICROSOFT AZURE

Azure Container Apps

Compute — 3 services

Azure API Management

Secure entry / edge

Azure AI Foundry

The AI recovery & messaging agents

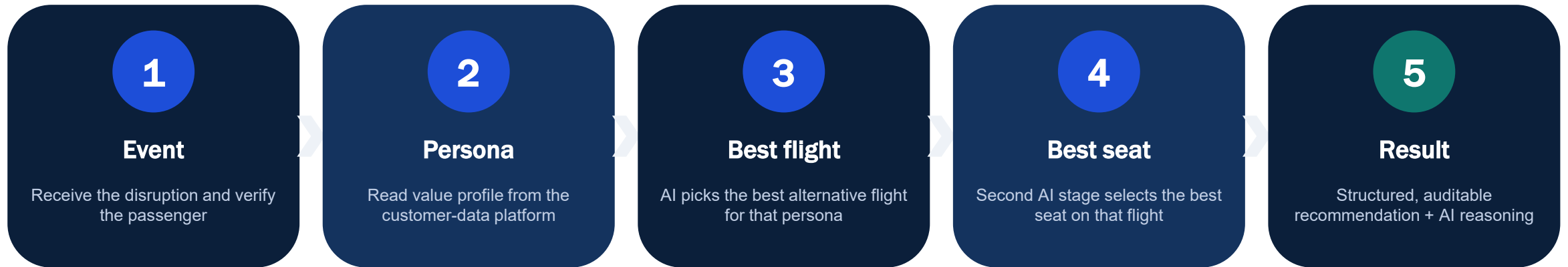
Azure Cosmos DB

Per-tenant configuration

Azure Managed Redis

Caching & precompute acceleration

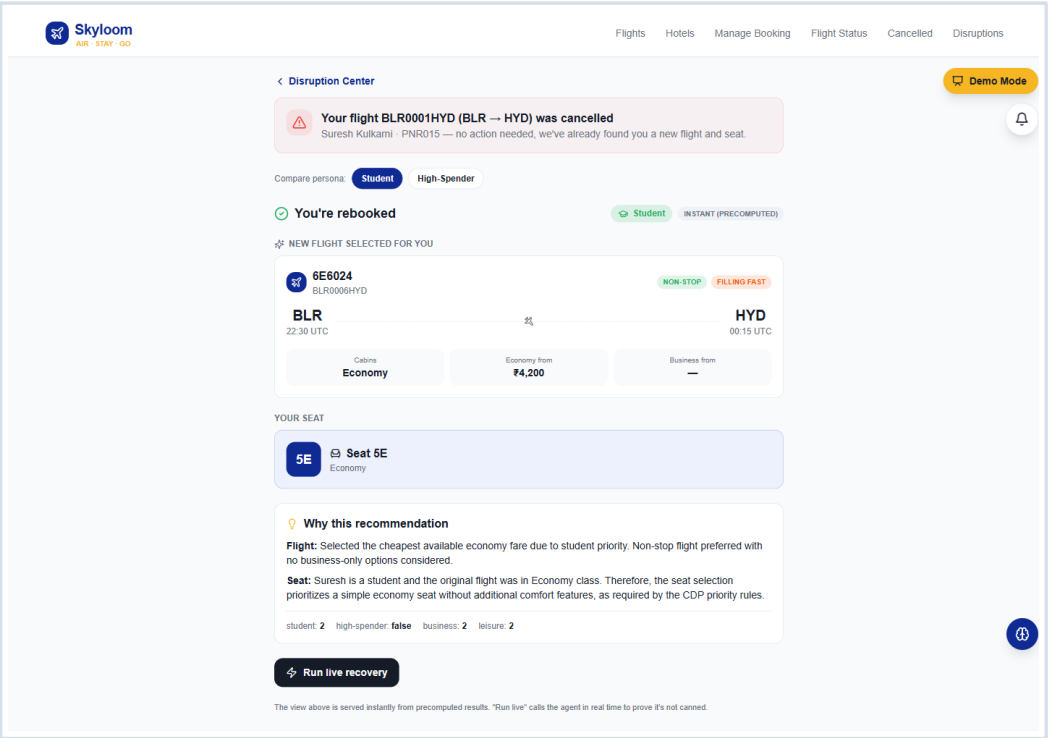
End-to-end recovery flow



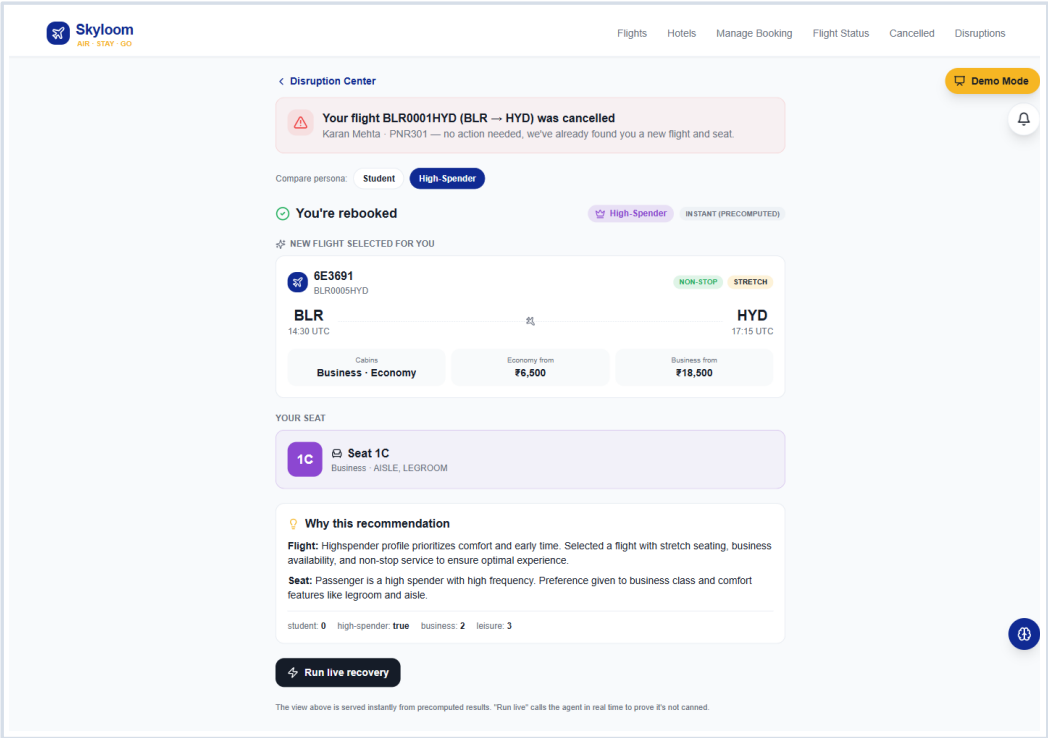
Two-stage design (flight → seat) cut redundant data fetches from many-per-request to one — improving speed and binding the seat to the chosen flight. *(Source: changelog 2026-05-30)*

Same disruption, different right answer

Same cancelled flight (BLR0001HYD, BLR→HYD). The only difference is passenger value — and the platform acts on it automatically.



STUDENT · PNR015 — cheapest economy ₹4,200, seat 5E (Y)



HIGH-SPENDER · PNR301 — business ₹18,500, seat 1C (C) aisle/legroom

Identical disruption & inventory → personalised, auditable outcomes with plain-language AI reasoning. *Source: live demo, 2026-05-31*

Whole-flight recovery in one pass

- A real disruption is an entire flight — `prodDisruption` discovers all affected passengers and recovers them in a single AI pass.

- Personas preserved at scale: students kept on cheapest economy, high-value flyers upgraded to business.

- Single-pane ops view: status per passenger, persona mix, rebooked-cabin split, and exceptions needing a human.

- Honest note: seat assignment depends on inventory; constrained inventory routes some passengers to an agent. (*known demo-data limitation*)



[Flights](#)
[Hotels](#)
[Manage Booking](#)
[Flight Status](#)
[Cancelled](#)
[Disruptions](#)

Disruption Center

Operations view · live disruption recovery

Ops Dashboard

Demo Mode





BLR0001HYD
CANCELLED

Bengaluru (BLR) → Hyderabad (HYD) · Operational cancellation


6

passengers impacted

Run bulk recovery →

Impacted passengers

6/6 recovery plans ready

PASSENGER	PERSONA	NEW FLIGHT	SEAT	FARE
Suresh Kulkarni PNR015	 Student	BLR0006HYD Economy · non-stop	5E	₹4,200 >
Karan Mehta PNR301	 High-Spender	BLR0005HYD Business · non-stop	1C	₹18,500 >
Nair PNR018	 Student	BLR0006HYD Economy · non-stop	5E	₹4,200 >
Yadav PNR017	 Student	BLR0005HYD Business · non-stop	1A	₹18,500 >
Rao PNR016	 Student	BLR0006HYD Economy · non-stop	5B	₹4,200 >
Yadav PNR002	 Student	BLR0006HYD Economy · non-stop	5B	₹4,200 >

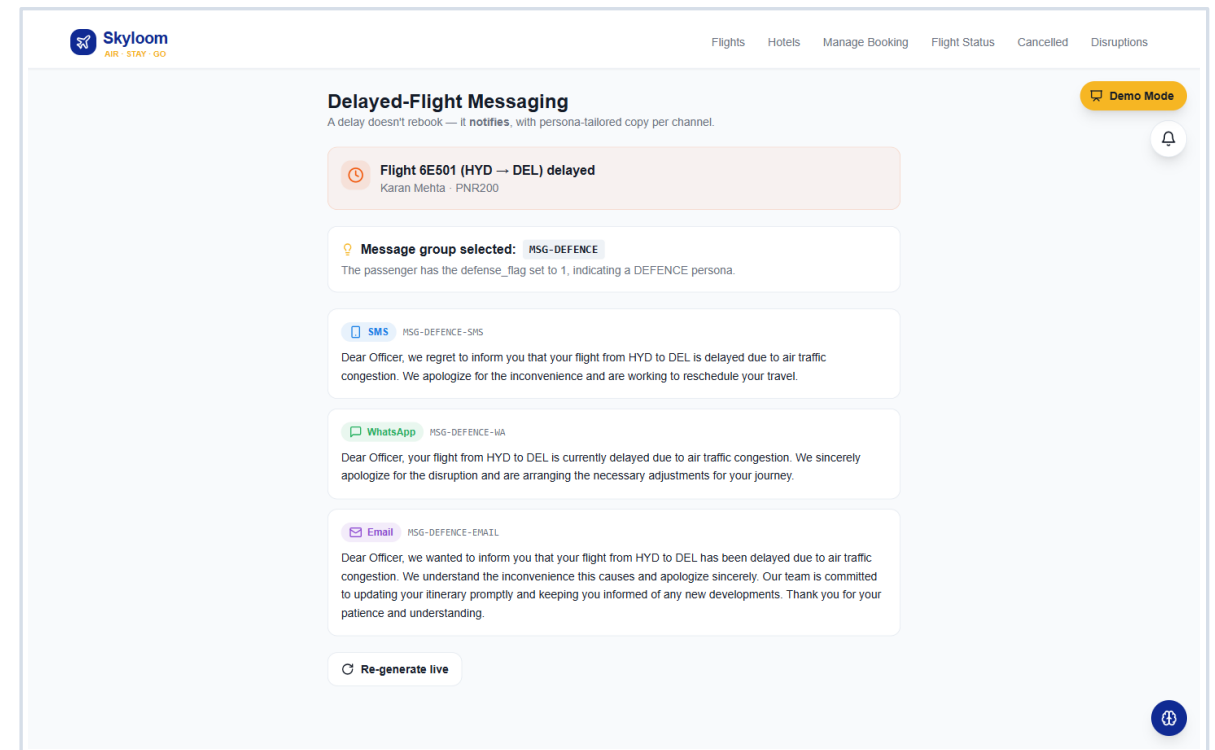
Each passenger gets a persona-personalized flight + seat. Click any row to see the passenger-facing recovery and reasoning.



Disruption Center — whole-flight view: persona → flight → seat → fare (ops dashboard companion view also available)

Persona-tailored, multi-channel notification

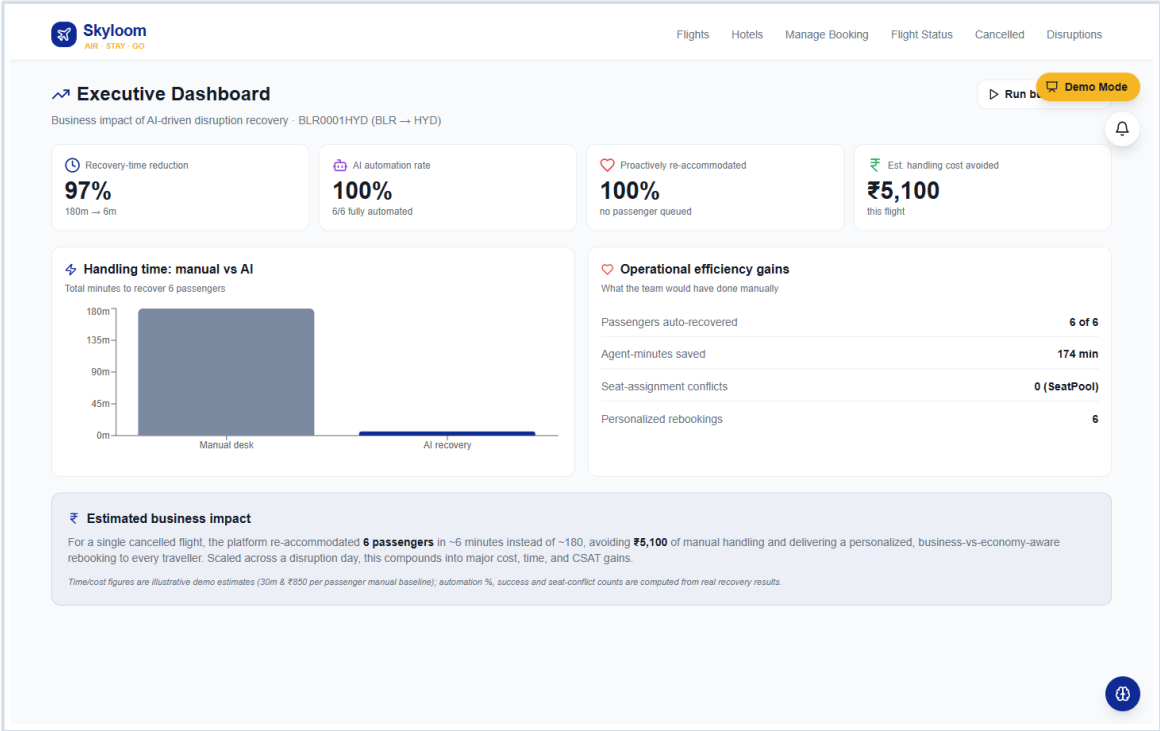
- Delays don't rebook — they notify. The AI selects the right message group for the persona and generates SMS, WhatsApp and Email copy.
- Example (PNR200, Defence persona): all three channels appropriately addressed, with delay, apology and next steps. *(Project: live demo)*
- Consistent, on-brand, persona-aware communication at the moment of disruption — the proactive certainty customers value most. *(Industry: McKinsey)*
- Message content is generated; actual dispatch (telco / email gateways) is a deliberate next-step integration.



Delayed-flight messaging — persona-tailored SMS / WhatsApp / Email

Faster, cheaper, consistent — and personalised

- Speed: passenger-facing recovery served in under a second from precompute, vs. minutes of manual triage. *(measured 0.26s)*
- Automation: the cancelled-flight demo auto-recovered passengers with zero manual escalations, freeing agents for exceptions. *(live bulk run)*
- Illustrative model: at a ₹850 / 30-min manual baseline per passenger, one cancelled flight avoids ~₹5,100 of handling and ~174 agent-minutes — compounding across a disruption day. *(Illustrative estimate)*
- Customer experience: faster, consistent, personalised recovery is linked to higher satisfaction and lower contact volume. *(Industry: McKinsey)*
- We do not yet have production cost-saved or NPS figures from a live airline — those are the rollout's success metrics.



Executive view — figures labelled illustrative; automation & success counts from real runs

BLR0001HYD cancellation — see it for real

- 1

Personalisation

Student → cheapest economy vs high-spender → business, same flight
- 2

Bulk recovery

Recover the whole flight in one pass; personas spread, distinct seats
- 3

Instant serve

Open a passenger's precomputed recovery in under a second
- 4

Delay messaging

A delayed flight generates persona-tailored SMS / WhatsApp / Email

Skyloom

AIR · STAY · GO

Flights

Hotels

Manage Booking

Flight Status

Cancelled

Disruptions

List of Cancelled Flights

Live from disruption records · 5 cancelled.

Demo Mode

BLR0001HYD

CANCELLED

LIVE

BLR → HYD

6 passengers impacted — AI recovery available

Open Disruption Center →

Enter Flight Number

Enter Sector (e.g. BLR - HYD)

Search

Flight	Sector	Impacted	Status
BOM0001GOI	BOM - GOI	1 passenger	Cancelled
DEL0019BLR	DEL - BLR	1 passenger	Cancelled
DEL0011BLR	DEL - BLR	1 passenger	Cancelled
CCU0001DEL	CCU - DEL	1 passenger	Cancelled
Delayed flights			
HYD0041DEL	HYD - DEL	5 passengers	Delayed

Scenario: BLR0001HYD (Bengaluru→Hyderabad) cancelled · 6 impacted passengers, from live disruption records. Agents execute on Azure AI Foundry.

Live disruption records — real Cosmos data via the platform

Validated core · target-state controls, clearly distinguished

EDGE & SECURITY — architected / pre-production hardening

Azure API Management

JWT auth · rate limit

Private Endpoints

No public data plane

Managed Identity

No secrets in code

Azure Key Vault

Secret store

Multi-region DR

Front Door failover

Azure Container Apps

Public API · Admin backend · Internal MCP orchestration

Azure AI Foundry

Recovery & Messaging agents · per-tenant name+version · hot-swappable

Azure Cosmos DB

Per-tenant configuration · agents · prompts · templates

Azure Managed Redis

Cache + precompute · ~6–7× warm speedup (measured cross-region)

■ Implemented & validated

■ Architected / target-state

Multi-tenant isolation by `tenant_id` across every layer · no passenger PII persisted (in-memory only).

A clear path from validated capability to live operations

NOW

- Performance pass — connection pooling / caching to cut live-recovery latency (~31–51s today)
- Turn Redis acceleration on by default
- Tighten demo data / seat inventory

NEXT

- Live-airline integration — replace mock providers
- Endpoint authentication + CORS lockdown
- Onboard additional tenants (config-only)

LATER

- Async job queue + worker pool
- Multi-region active-active
- SaaS Accelerator + Azure Marketplace metered billing

KEY CONSIDERATIONS (HONEST)

Currently mock-backed and single-tenant-wired · live on-demand recovery ~31–51s (mitigated by precompute + the performance pass) · no production authentication yet. These are normal pre-production items with defined owners and mitigations — not blockers to the validated capability.

What we're asking leadership for

1

Endorse a live-airline pilot

Move from validated capability to one design-partner airline

2

Approve the hardening sprint

Performance pass · authentication · tenant onboarding

3

Align on pilot success metrics

Recovery time · automation rate · contact deflection · CSAT / NPS

4

Fund the enterprise roadmap

Async · multi-region · Azure Marketplace

Immediate action: stand up a 1–2 tenant pilot environment and instrument the business KPIs.